Expectation

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First, an example

Roll a fair six-sided die and let X be the result. Each face has probability 1/6, so the expectation is

$$E[X] = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$

Read aloud: "The expectation of X equals one plus two plus three plus four plus five plus six, all over six, which equals twenty-one over six, which equals three point five." Note that 3.5 is *not* a possible outcome of the die: the expectation is the long-run average, not a typical value. Compare also

$$E[X^2] = \frac{1+4+9+16+25+36}{6} = \frac{91}{6} \approx 15.17, \qquad \big(E[X]\big)^2 = (3.5)^2 = 12.25.$$

Read aloud: "The expectation of X squared equals ninety-one over six, approximately fifteen point one seven, while the square of the expectation of X is twelve point two five." Since $x \mapsto x^2$ is convex, Jensen's inequality predicts $E[X^2] \geq \big(E[X]\big)^2$, and indeed $15.17 > 12.25$.

1 Definitions

Definition 1 (Expectation, discrete)**.** *Let X be a discrete random variable on a probability space $(\Omega, \mathcal{F}, P)$ taking values $\{k_i\}_{i\geq 1} \subset \mathbb{R}$ with probability mass function $p_X(k) = P(X = k)$. If $\sum_i |k_i|\, p_X(k_i) < \infty$, the* expectation *of X is*

$$E[X] \;=\; \sum_i k_i\, p_X(k_i).$$

Read aloud: *"The expectation of X equals the sum over i of k-sub-i times p-sub-X of k-sub-i."*

Definition 2 (Expectation, continuous)**.** *Let X be a continuous random variable with probability density function f_X. If $\int_{-\infty}^{\infty} |x|\, f_X(x)\, dx < \infty$, then*

$$E[X] \;=\; \int_{-\infty}^{\infty} x\, f_X(x)\, dx.$$

Read aloud: *"The expectation of X equals the integral from minus infinity to infinity of x times f-sub-X of x, dx."*

Definition 3 (Expectation, Lebesgue)**.** *For an integrable measurable $X : \Omega \to \mathbb{R}$,*

$$E[X] \;=\; \int_\Omega X(\omega)\, dP(\omega).$$

Read aloud: *"The expectation of X equals the integral over Omega of X of omega, with respect to dP of omega." This is constructed first for simple functions, extended to non-negative measurable functions via monotone approximation, and to signed functions via $X = X^+ - X^-$.*

2 Linearity of Expectation

Theorem 1 (Linearity of Expectation)**.** *Let X and Y be random variables on a common probability space with $E[|X|] < \infty$ and $E[|Y|] < \infty$, and let $a, b \in \mathbb{R}$. Then*

$$E[aX + bY] \;=\; a\,E[X] + b\,E[Y].$$

Read aloud: *"The expectation of a X plus b Y equals a times the expectation of X plus b times the expectation of Y." Linearity holds whether or not X and Y are independent.*

Proof (discrete case). Assume X and Y are discrete random variables with joint pmf $p_{X,Y}(x,y) = P(X = x, Y = y)$, and that $\sum_{x,y} |x|\, p_{X,Y}(x,y) < \infty$ and likewise for Y. Let $Z = aX + bY$.

By LOTUS applied to the joint pmf,

$$\begin{aligned} E[Z] \;=\; E[aX + bY] &= \sum_x \sum_y (ax + by)\, p_{X,Y}(x,y) \\ &= a \sum_x \sum_y x\, p_{X,Y}(x,y) \;+\; b \sum_x \sum_y y\, p_{X,Y}(x,y). \end{aligned}$$

Read aloud: "The expectation of Z equals a times the double sum over x and y of x times the joint pmf, plus b times the double sum over x and y of y times the joint pmf." Absolute convergence (from the integrability assumptions) permits us to rearrange the double sum and apply Fubini for series. Marginalising the joint pmf gives $\sum_y p_{X,Y}(x,y) = p_X(x)$ and $\sum_x p_{X,Y}(x,y) = p_Y(y)$, so

$$\begin{aligned} E[Z] &= a \sum_x x \left(\sum_y p_{X,Y}(x,y) \right) + b \sum_y y \left(\sum_x p_{X,Y}(x,y) \right) \\ &= a \sum_x x\, p_X(x) + b \sum_y y\, p_Y(y) \\ &= a\,E[X] + b\,E[Y]. \end{aligned}$$

Read aloud: "Marginalising the joint pmf collapses each inner sum to the marginal pmf, giving a times the expectation of X plus b times the expectation of Y." No use of independence was required: the joint pmf $p_{X,Y}$ carries any dependence structure, and it factors out of each single-variable sum only through its marginals. □

Proposition 1 (Jensen's inequality)**.** *Let $\phi : \mathbb{R} \to \mathbb{R}$ be convex and let X be an integrable random variable with $E[|\phi(X)|] < \infty$. Then*

$$\phi\big(E[X]\big) \;\le\; E\big[\phi(X)\big].$$

Read aloud: *"Phi of the expectation of X is less than or equal to the expectation of phi of X."*

Proposition 2 (LOTUS)**.** *For measurable $g : \mathbb{R} \to \mathbb{R}$ such that $g(X)$ is integrable,*

$$E[g(X)] \;=\; \int_{\mathbb{R}} g(x)\, f_X(x)\, dx \quad \textit{(continuous)}, \qquad E[g(X)] \;=\; \sum_k g(k)\, p_X(k) \quad \textit{(discrete)}.$$

Read aloud: *"The expectation of g of X equals the integral over the reals of g of x times the density of X, dx, in the continuous case, and the sum over k of g of k times the pmf of X at k in the discrete case."*

3 Examples

Bernoulli. If $X \sim \text{Bernoulli}(p)$, then $E[X] = 1 \cdot p + 0 \cdot (1-p) = p$.

Uniform $[a, b]$. If $X \sim \text{Uniform}[a, b]$, then

$$E[X] = \int_a^b \frac{x}{b-a}\, dx = \frac{a+b}{2}.$$

Read aloud: "The expectation of X equals the integral from a to b of x over b minus a, dx, which equals a plus b over two."

Binomial. If $X \sim \text{Binomial}(n, p)$, write $X = \sum_{i=1}^{n} X_i$ with $X_i \sim \text{Bernoulli}(p)$ iid. By Theorem 1, $E[X] = \sum_{i=1}^{n} E[X_i] = np$.